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Roll No

MI-5002-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Mining Surveying - II

Time : Three Hours

Maximum Marks : 70

- Note:** i) Answer any five questions.
ii) All questions carry equal marks.

1. Attempt any four:
 - a) State the principle of tacheometric surveying?
 - b) Define the term Photogrammetry?
 - c) Define the terms Longitude and Latitude?
 - d) List the various correlation methods for surface and underground?
 - e) State the purpose of correlation?
 - f) List the fundamental measurement by Total Station?
2.
 - a) The stadia reading with horizontal sight on a vertical staff held at 50 m from a tacheometer are 1.285 and 1.780. The focal length of the object glass is 25 cm and the distance between the object glass and the vertical axis of the tacheometer is 15 cm. the stadia interval in mm?
 - b) Explain the terms Simple curve and Compound curve?

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3.
 - a) What is meant by transition curves? Why a transition curve is needed on roadways or railways? What conditions are to be full-filled by it? Give the advantage of such curve.
 - b) Explain how an underground road can be set out in simple circular curve. A road advancing on a bearing $N 35^{\circ}12' E$ is to be turned due North by means of a circular curve of radius 36 meters. Calculate the lengths and quadrantal bearing of the chords, necessary to set out the curve.
4. Enumerate main sources of errors in theodolite surveying and their prevention.
5.
 - a) Explain the different rules of area measurement of irregular boundaries.
 - b) Explain planimeter with neat sketch.
6.
 - a) Explain ediograph and pentagraph with neat sketch.
 - b) Three borehole A,B and C have been put down to a coal seam. B is due to north of A 1250 m and C is $N 60^{\circ}W$ 800 m from A. the surface levels of boreholes are the same. The depth of A = 750m, B = 1300m and C = 950m. Calculate the rate and direction of full dip of the seam.
7. Describe the method of correlation by direct traversing?
8.
 - a) Describe various symbols used in mine plans.
 - b) Explain how the measurements of height and distance of accessible and inaccessible points can be taken by a theodolite.
